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Reversed Future Drying-Wetting Precipitation Patterns Over the Northwestern and Southeastern East Asia Driven by Reduced Aerosol Emissions

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Key Points:

- Dry-wet contrast in East Asia has diminished due to increasing precipitation in dryland and decreasing precipitation in humid region
- Historical greenhouse gas and aerosol emissions dominated the precipitation trends in dryland and humid region of East Asia, respectively
- Future aerosol emission reductions are projected to reverse the historically weakened dry-wet contrast in East Asia via altered evaporation

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract

To investigate drying-wetting patterns of precipitation trends in East Asia, we examine the past and future precipitation changes and their underlying control factors using observations and CMIP6 simulations. Over the past seven decades, greenhouse gases enhanced evaporation through enhanced net longwave radiation, raising dryland precipitation, while aerosols suppressed evaporation by reducing net shortwave radiation, lowering humid region precipitation, thereby weakening the dry-wet contrast. Future CMIP6 projections show greater precipitation increases in humid region than in dryland under both intermediate- and high-emission scenarios. Observation-constrained projections indicate a slight decrease in dryland and an increase in humid region, strengthening the dry-wet contrast. This shift stems from aerosol reductions that increased net shortwave radiation, enhancing evaporation in humid region and reversing the historical suppressive effect of aerosols on precipitation. These findings highlight the distinct roles of greenhouse gases and aerosols in modulating regional precipitation, providing insights for climate adaptation strategies in East Asia.

Plain Language Summary

Precipitation is an important climate factor with significant effect on people's lives. This study uses multiple observational data sets and climate models to analyze changing patterns of precipitation in East Asia and factors affecting it. Over the past seven decades, greenhouse gases increased net longwave radiation, enhancing evaporation and leading to more precipitation in dryland, while aerosols reduced net shortwave radiation, suppressing evaporation and decreasing precipitation in humid region. Consequently, the dry-wet contrast in East Asia has diminished. Future projections from climate models show overall precipitation increases under intermediate- and high-emission scenarios, with significantly larger magnitude in humid region than dryland. The model projections constrained by observations show moderate increase in humid region and weak decrease in dryland, suggesting the dry-wet contrast will strengthen. This shift is primarily driven by anticipated aerosol emission reductions, increasing net shortwave radiation, thereby enhancing local evaporation and reversing the historically suppressive effect of aerosols on precipitation in humid region. These findings clarify the divergent roles of greenhouse gases and aerosols in regional precipitation and underscore the need for adaptive strategies amid projected strengthened dry-wet contrast in East Asia.

1. Introduction

Precipitation significantly impacts society, agriculture, and the environment (Pendergrass et al., 2017). Consequently, changes in precipitation patterns have received increasing attention. Previous studies have indicated a decline in global terrestrial monsoon precipitation since the 1950s (Wang & Ding, 2006). From 1980 to 2019, precipitation significantly increased in tropical Africa, Europe, eastern North America, and Central Asia, while it decreased notably in central South America, western North America, North Africa, and the Middle East (Gulev et al., 2021). Although annual precipitation has declined in tropical, subtropical, and mid-latitude regions, global land areas overall have experienced a slight increase in annual precipitation in recent decades (Alizadeh, 2023). Nonetheless, the increase or decrease in regional precipitation trends may vary by season. Taking Southwest Asia as an example: winter precipitation exhibits particularly pronounced decreases in southwest and eastern Iran as well as southwest Pakistan, while summer precipitation in this region remains relatively stable (Alizadeh & Babaei, 2022).

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Global drought frequency and intensity have increased significantly (Milly & Dunne, 2016), particularly in East Asia (Van der Schrier et al., 2011), where the spatial patterns of precipitation strongly influence drought occurrence (Mirabbasi et al., 2012). In the semi-arid regions of East Asia, precipitation has increased due to the westward expansion of the monsoon (Guan et al., 2021). However, within China, the weakened East Asian monsoon since the late 1970s has created a distinct "south-flood-north-drought" pattern (Yu & Zhou, 2007; Yu et al., 2004; Zhou et al., 2009), while the arid northwest region has experienced a transition to warmer and wetter conditions since 2000 (Dong et al., 2024; Li & Ma, 2018; Wu et al., 2022; Zhang et al., 2021). In eastern China, annual precipitation shows a clear spatial gradient, with increasing trends in the south and decreasing trends in the north (Zhai et al., 2005).

The Coupled Model Intercomparison Project (CMIP), initiated by the World Climate Research Program (WCRP), has been widely used to investigate present and future precipitation changes (Almazroui et al., 2020; Makula & Zhou, 2022; Rivera & Arnould, 2020; Zhou et al., 2014). The sixth CMIP (CMIP6; Eyring et al., 2016) has improved global precipitation simulations (Lun et al., 2021; Zhu & Yang, 2021). For example, CMIP6 models reasonably simulate the spatial distribution of monsoon precipitation in East Asia (Jiang et al., 2023), and reproduce the patterns of annual mean precipitation in China (Tian et al., 2021). However, there are multiple uncertainties in model projections (Lehner et al., 2020). The CMIP6 models tend to significantly overestimate the rate of annual precipitation increase in Asia (Chai et al., 2022). Long-term projections of precipitation over the East Asian monsoon region are systematically overestimated under high-emission scenarios (Chen et al., 2020). Moreover, historical mean precipitation in China is generally overestimated by the CMIP6 models, suggesting persistent and potentially increasing uncertainties in future precipitation projections (Jia et al., 2023).

Under global warming, the precipitation pattern in East Asia is undergoing changes, yet CMIP6 simulations still present the aforementioned regional uncertainties. To tackle these uncertainties, this study analyzes the evolution characteristics of dry and wet patterns in key regions of East Asia by leveraging CMIP6 multi-model simulations and observational data. It quantifies the impacts of aerosols, greenhouse gases, and natural forcing on precipitation patterns. An observational constraint method is used to improve the CMIP6 projection of precipitation trend. The method constrains the dominant spatiotemporal modes of the precipitation based on the Empirical Orthogonal Function (EOF) analysis, which can filter out the noise from lesser spatiotemporal modes and capture the key signals influencing the regional precipitation trend. Through moisture and energy budget analysis, the mechanisms driving precipitation changes are further examined. These analytical approaches enable us to enhance the precipitation projections in East Asia and deepen our understanding of the regional precipitation variation mechanism. Thus, this study provides support for regional climate risk assessment and the formulation of adaptation strategies.

2. Data and Methods

2.1. Observational Data

Three data sets of monthly observational precipitation are employed in this study. Since CMIP6 model data span from 1850 to 2014, we select observational data sets with longer temporal coverage. The unselected data sets generally begin in 1979. Based on the above considerations, we select the following three data sets: the ERA5 data set from 1940 to 2022, provided by the European Center for Medium-Range Weather Forecasts (ECMWF) at a resolution of $1° \times 1°$ (Soci et al., 2024), the version 4.07 of the Climatic Research Unit (CRU) data set spanning 1901 to 2022 with a $0.5° \times 0.5°$ resolution (Harris et al., 2020), and the Precipitation Reconstruction over Land (PREC/L) data set from 1948 to 2023, provided by the National Oceanic and Atmospheric Administration (NOAA) at a resolution of $0.5° \times 0.5°$ (Chen et al., 2002). To illustrate the multi-data mean, all data sets are regridded to a uniform $1° \times 1°$ resolution using bilinear interpolation. The observational results presented here are based on the average of these three selected data sets, enhancing the reliability and representativeness of the results.

2.2. CMIP6 Simulations

This study uses precipitation data from 14 CMIP6 models, as listed in Tables S1 and S2 in Supporting Information S1. Models are selected based on the availability of data from four historical and two future experiments. The historical experiments include an all-forcing simulation (HIST) and three single-forcing simulations: greenhouse gases (Hist-GHG), natural forcing (Hist-NAT), and aerosols (Hist-AER) (Gillett et al., 2016). The historical simulations cover the period from 1850 to 2014, with analysis focused on 1948–2014 to match observational data availability. The future simulations include two Shared Socioeconomic Pathways: an intermediate-emission scenario (SSP2-4.5) and a high-emission scenario (SSP5-8.5) (O'Neill et al., 2017). The future simulations cover the period from 2015 to 2100. To attribute future precipitation changes, four models are selected from the initially 14 models based on their inclusion of three single-forcing simulations under future intermediate-emission scenario: greenhouse gases (SSP2-4.5-GHG), natural forcing (SSP2-4.5-NAT), and aerosols (SSP2-4.5-AER), as shown in Table S1 (Gillett et al., 2016). The future single-forcing simulations cover the period from 2021 to 2100. Note that the high-emission scenario does not include any single-forcing experiments. Six key variables are examined across seven selected models to analyze the drivers of precipitation changes. These seven models are chosen from the original 14 based on criteria detailed in the Methods section (see Tables S3 and S4 in Supporting Information S1). The mechanisms underlying evaporation changes are analyzed through key energy budget components, with detailed data presented in Table S5 in Supporting Information S1. Additionally, the aerosol optical depth at 550 nm from six of the seven models is analyzed to characterize aerosol emissions under both historical and future scenarios, but BCC-CSM2-MR lacks this variable. The resolution of each model is changed to $1° \times 1°$ using bilinear interpolation, and ensemble members of the same model in the group of experiments are averaged using an equal-weighted arithmetic mean (ensemble mean within the model).

2.3. Observational Constraint

Step 1: Model selection

To evaluate the accuracy of the spatial distribution of simulated precipitation across different models, the spatial correlation coefficients between 14 models and observational data are calculated to determine the degree to which each model reproduced the observational precipitation pattern (Taylor, 2001), and the seven models with correlation coefficients above the median are selected, identifying the models most strongly correlated with the observations (Figure S1 in Supporting Information S1). The ensemble mean precipitation anomalies of these selected models exhibit consistent patterns of precipitation change with those derived from the full fourteen-model ensemble during both historical and future periods (Figure S2 in Supporting Information S1).

Step 2: Extract the leading mode of precipitation variation

The dominant modes of precipitation are extracted using the EOF analysis (North et al., 1982), which decomposes precipitation anomalies into spatial patterns and principal component (PC) time series (Figure S3 in Supporting Information S1). The first seven EOF modes account for 96% of the observed variance and 98% of the modeled variance (Figure S4 in Supporting Information S1). Since the remaining modes contribute minimally, these leading modes effectively represent the dominant variability in the precipitation. Additionally, based on the North test (North et al., 1982), the eigenvalues obtained from EOF analysis indicate that the first seven modes are all statistically significant.

Step 3: Constrain the PC time series of leading modes using observation

The observational constraints on the leading EOF modes are utilized to effectively constrain the leading modes with signals and mitigate the interference of noise from minor modes. This approach has been successfully used in seasonal forecasting and correcting the spatial coverage of drylands (Huang et al., 2016; Kang et al., 2004; Kim et al., 2004; Lim et al., 2011). Thus, we employ the method to correct the precipitation simulated by CMIP6 models, as detailed below.

To correct model simulations during the historical period, each model's precipitation field is projected onto the observational EOF spatial modes to derive the corresponding PC time series. To quantify the correlation between model simulations and observed PCs, this study employs the cross-validation linear regression method (Feddersen et al., 1999). The data from one target year are excluded at a time, and a linear regression model is built using the remaining years. The data of the excluded year is then predicted using the regression model. Specifically, a linear regression relationship between the simulations and the observations is established using data from all other years, allowing for the prediction of PC time series for the excluded years. This process is repeated for all years, ultimately generating multiple adjusted PC time series. The adjusted time series are multiplied by the observed spatial modes, and the results are combined mode by mode to obtain the estimated annual precipitation for each model. The Anomaly Correlation Coefficient (ACC) is calculated to determine the optimal number of leading modes, which is seven as detailed below. The corrected historical precipitation is then reconstructed by summing the contributions of these first seven modes (Huang et al., 2016).

To correct future projection of precipitation, the models under the SSP2-4.5 and SSP5-8.5 scenarios are projected onto the observational spatial EOF modes, yielding the corresponding PC time series for each model. The regression coefficients and intercepts derived from the historical period are then applied to adjust the PC time series for each future scenario. The adjusted PC time series are multiplied by the observed spatial modes, and the precipitation under the first seven modes is summed to obtain the corrected annual precipitation for each model under its respective scenario (Figure S5 in Supporting Information S1). The consistency of the leading EOF modes of precipitation between historical and future periods is examined by calculating spatial correlation coefficients of the EOF modes among the HIST, SSP2-4.5, and SSP5-8.5 scenarios. Results indicate that the persistence of leading EOF modes into the future varies among different models. The mean spatial correlation coefficient is 0.35, with values ranging from 0.05 to 0.62 (Figure S6 in Supporting Information S1). To further investigate the source of such inconsistency, we analyzed the first EOF mode under three single-forcing historical experiments: Hist-GHG, Hist-AER, and Hist-NAT. The results show that spatial patterns of EOF under different forcing experiments are different, leading to distinct precipitation variations under each individual forcing (Figure S7 in Supporting Information S1). The combined influence of these forcings accounts for the overall variability in the HIST simulation. Thus, the inconsistent EOF modes across models are probably caused by differences in modeled precipitation responses to external forcings.

The ACC is a widely employed indicator in the field of climate prediction for evaluating the consistency between simulated results from climate models and observational data. ACC is defined as follows:

$$\mathrm{ACC} = \frac{\sum_{i=1}^{n}\left(F_i - C_i - M_i^{F,C}\right)\left(O_i - C_i - M_i^{O,C}\right)\cos\varphi_i}{\sqrt{\sum_{i=1}^{n}\left(F_i - C_i - M_i^{F,C}\right)^2\cos\varphi_i}\sqrt{\sum_{i=1}^{n}\left(O_i - C_i - M_i^{O,C}\right)^2\cos\varphi_i}} \tag{1}$$

where n is the number of spatial grid points, φ_i is the latitude of grid point i, F is the prediction, O is the observed value, and C is the climatological mean, $M_i^{F,C} = \frac{1}{n}\sum_{i=1}^{n}(F_i - C_i)$, $M_i^{O,C} = \frac{1}{n}\sum_{i=1}^{n}(O_i - C_i)$. First, the annual ACC is calculated between each model field and the observed field, followed by averaging all the results to obtain a mean ACC for a period of 67 years. The relationship between mean ACC and the number of different leading PCs is shown in Figure S8 in Supporting Information S1. A decline in ACC is observed when the number of leading PCs exceeds seven. Therefore, retaining the seven leading PCs is considered the optimal choice for the number of leading modes (Figure S8 in Supporting Information S1).

2.4. Moisture and Energy Budget Equations

The atmospheric moisture budget equation is adopted to identify the primary processes driving precipitation trend (Chou et al., 2009; Luo et al., 2025), written as:

$$P' = E' - \left\langle \omega' \frac{\partial \overline{q}}{\partial p} \right\rangle - \langle \boldsymbol{V}' \cdot \nabla \overline{q} \rangle - \left\langle \overline{\omega} \frac{\partial q'}{\partial p} \right\rangle - \langle \overline{\boldsymbol{V}} \cdot \nabla q' \rangle + Res \tag{2}$$

where P, E, ω, q, $\boldsymbol{V}$, and p denote precipitation, evaporation, vertical velocity, specific humidity, horizontal wind, and pressure, respectively. The primes denote the monthly anomaly, and the overbars denote the climatological mean. The second to fifth terms on the right-hand side correspond to the dynamic terms related to vertical convection and horizontal advection, and the thermodynamic terms associated with vertical convection and horizontal advection, respectively. Residuals (Res) encompass the nonlinear high-frequency eddy component and errors arising from the finite differential method. The symbol $\langle\ \rangle$ represents the vertical integration from the surface to 300 hPa, shown as:

$$\langle X \rangle = -\frac{1}{g}\int_{p_s}^{p_T} X dp \tag{3}$$

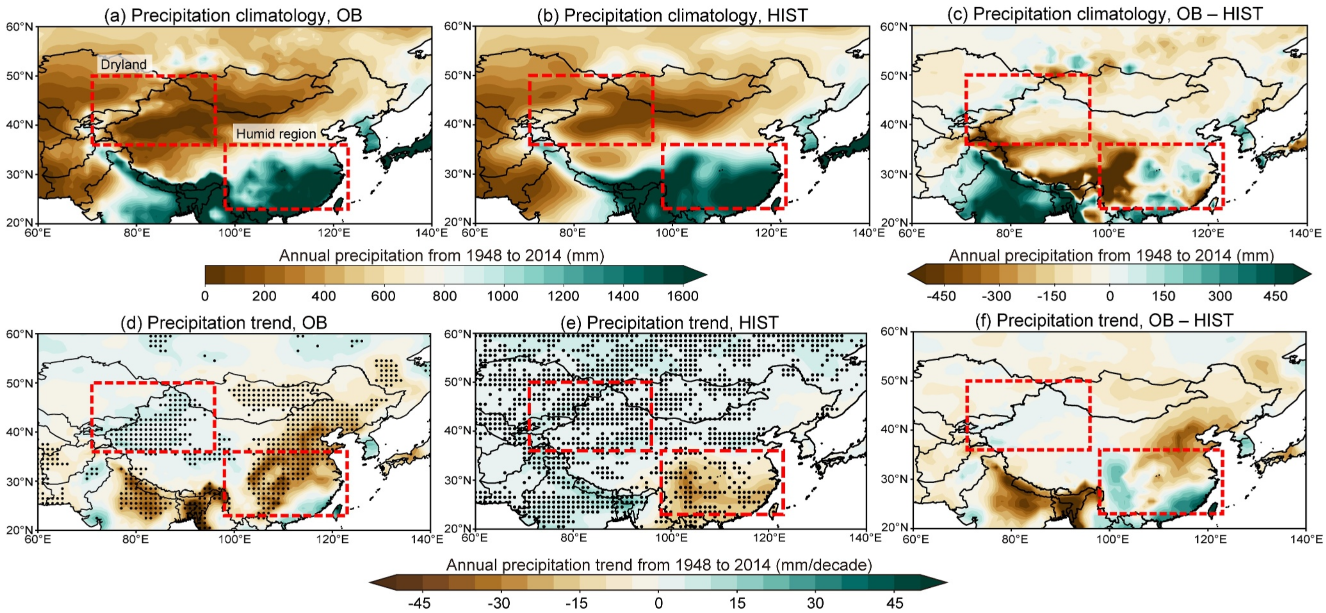

Figure 1. (a), (b) The climatological mean of annual precipitation from 1948 to 2014 is calculated as the average of three observational data sets for observations (OB), and as the multimodel ensemble mean for CMIP6 historical simulations (HIST), respectively. The two rectangles outline the domains examined next. (d), (e) Linear trends in annual precipitation in OB and HIST from 1948 to 2014, respectively. The dots in (d) indicate significant trends at the 95% confidence level based on the two-tailed Student's *t*-test, while the dots in (e) indicate both the 95% statistical significance of trends and the agreement in models exhibiting the same sign as the ensemble mean trend across at least nine out of 14 models. (c), (f) The difference in climatology and trend, which is calculated as the precipitation in OB minus that in HIST, respectively.

where g is the acceleration due to gravity, p_S is surface pressure and p_T here is 300 hPa.

The key processes driving changes in evaporation can be identified using the energy budget equation (Xie et al., 2023), written as:

$$E = \frac{1}{L_v}(\mathrm{SW_{net}} + \mathrm{LW_{net}} - \mathrm{SH}) \tag{4}$$

where $\mathrm{SW_{net}}$, $\mathrm{LW_{net}}$, and SH indicate net shortwave radiation, net longwave radiation and sensible heat flux, respectively. $\mathrm{SW_{net}}$ and $\mathrm{LW_{net}}$ are downward positive, while SH is upward positive. L_v denotes the latent heat of vapourization of water, approximately 2.5×10^6 J/kg.

2.5. Statistical Methods

The precipitation trend is estimated using linear regression, and its statistical significance is evaluated at the 95% confidence level by two-tailed Student's *t*-test. The Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN) approach is used to decompose the model and observational time series of precipitation (Torres et al., 2011), and the last two Intrinsic Mode Functions (IMFs) from the decomposition are added together to study the interdecadal to long-term variation of precipitation. This method is capable of reducing the interference of interannual variability on time series analysis and observational constraints. However, the results illustrated in Figures 1, 2c–2h, and 4c–4j, Figures S1, S5, S6, and S10–S23 in Supporting Information S1 rely on the raw data without CEEMDAN processing. To evaluate CEEMDAN's applicability to trend analysis, we additionally applied CEEMDAN to derive spatial trends. The results indicate that the trends derived from CEEMDAN-processed data are highly consistent with those obtained from the raw data, confirming the robustness of our trend-based results (Figures 1d, 1e and Figure S9 in Supporting Information S1).

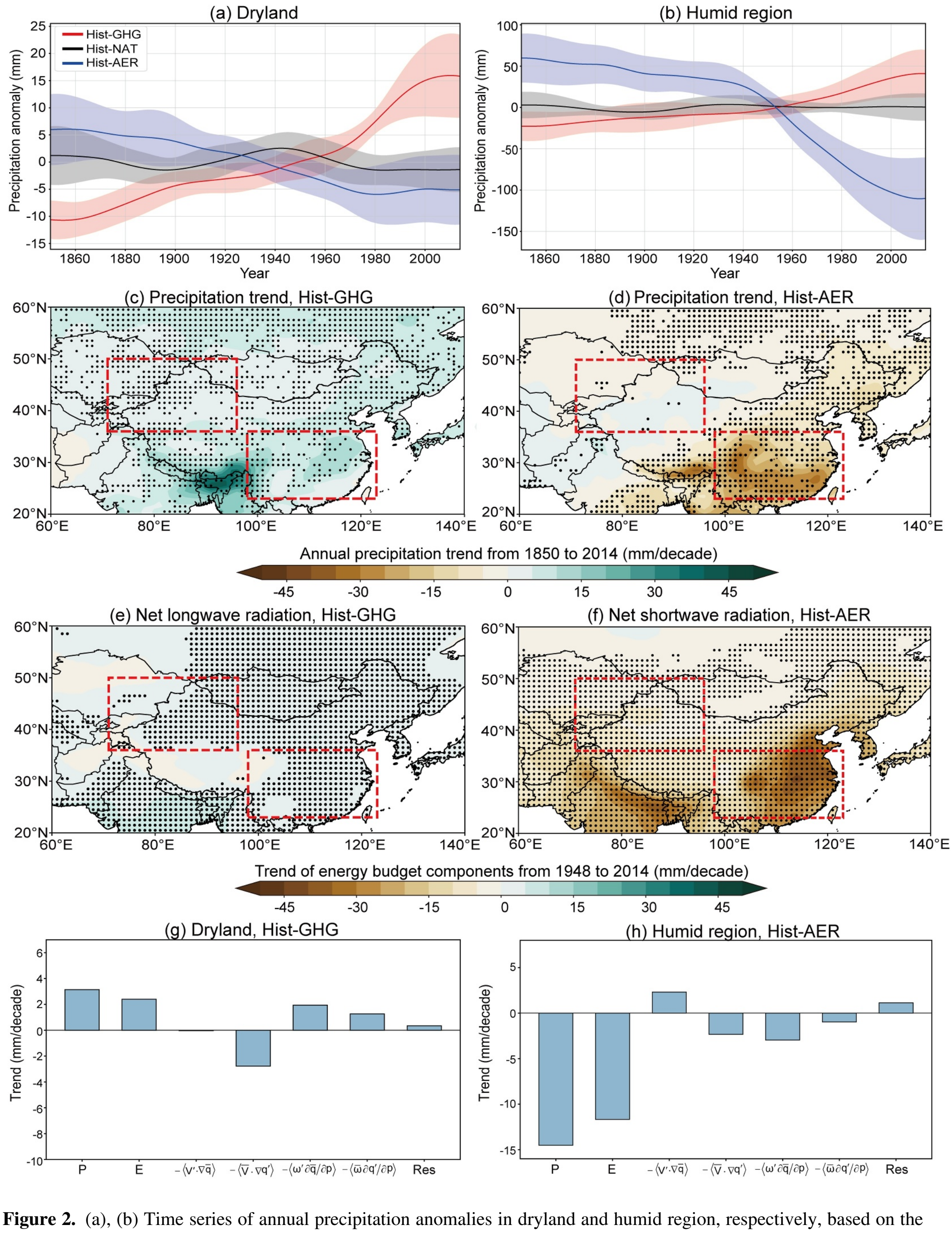

Figure 2. (a), (b) Time series of annual precipitation anomalies in dryland and humid region, respectively, based on the multimodel ensemble mean of different CMIP6 forcing experiments from 1850 to 2014. The anomalies are calculated for each model with respect to the 1850 to 2014 average. The shaded area is an uncertainty range of one standard deviation across different models. Hist-NAT represents natural forcing. (c), (d) Linear trends in annual precipitation from 1850 to 2014 under greenhouse gas forcing (Hist-GHG) and aerosol forcing (Hist-AER), respectively. The dots indicate significant trends at the 95% confidence level based on the two-tailed Student's *t*-test and the agreement in models exhibiting the same sign as the ensemble mean trend across at least nine out of 14 models. (e), (f) Linear trends in net longwave radiation in the Hist-GHG experiment and net shortwave radiation in the Hist-AER experiment, respectively. The dots indicate significant trends at the 95% confidence level based on the two-tailed Student's *t*-test. (g), (h) Terms of the moisture budget equation for dryland in the Hist-GHG experiment and humid region in the Hist-AER experiment. The CEEMDAN method is used to produce the time series in (a) and (b).

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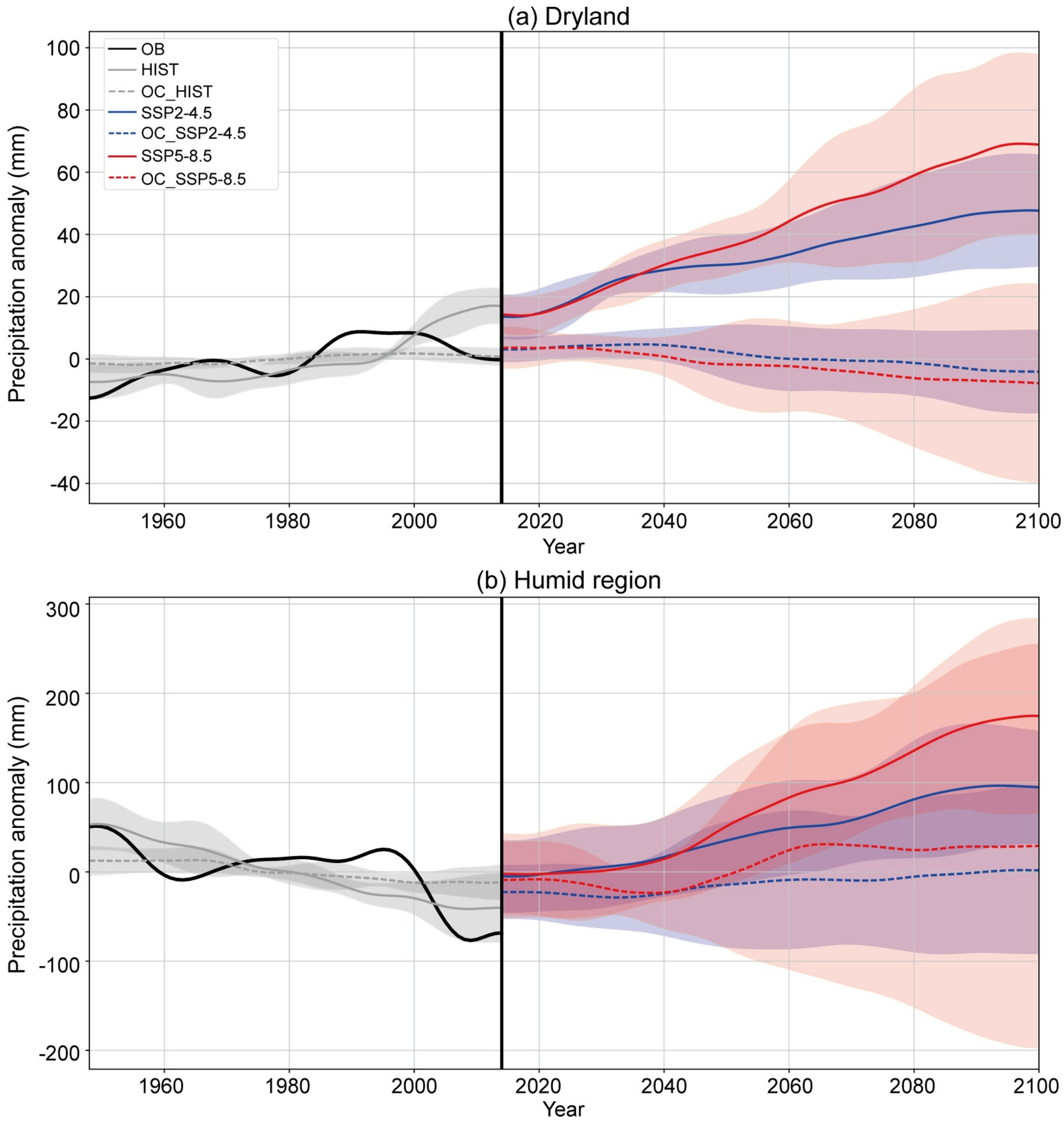

Figure 3. The time series of annual precipitation anomalies after observational constraints (OC). (a), (b) Time series of annual precipitation anomalies in the dryland and humid region, respectively, based on the multimodel ensemble mean of CMIP6 for the period from 1948 to 2100. Historical and future anomalies for each model are calculated relative to the 1948–2014 average. The shaded area is an uncertainty range of one standard deviation across different models. The CEEMDAN method is used to produce the time series.

3. Results

3.1. Historical Precipitation Trends in East Asia and Their Causes

The distribution of historical precipitation in simulations in East Asia is generally consistent with the observations, characterized by dry conditions in the northwest region and humid conditions in the southeast region. Although the simulation results slightly overestimate precipitation in the west of humid region compared to the observations, overall CMIP6 demonstrates a high capability in simulating East Asian precipitation (Figures 1a and 1b). When comparing the linear trends of historical simulations and the observations, precipitation decreases in the majority of the humid region while increases in dryland (Figures 1d and 1e). However, in the model simulations, notable deviations are observed in the depiction of precipitation characteristics over regions with complex topography, like the Tibetan Plateau (Figures 1c and 1f). These differences highlight the limitations of the model in capturing the precipitation characteristics of complex terrains. These results indicate that the humid region is gradually becoming drier while dryland is gradually becoming wetter, so the dry-wet contrast between

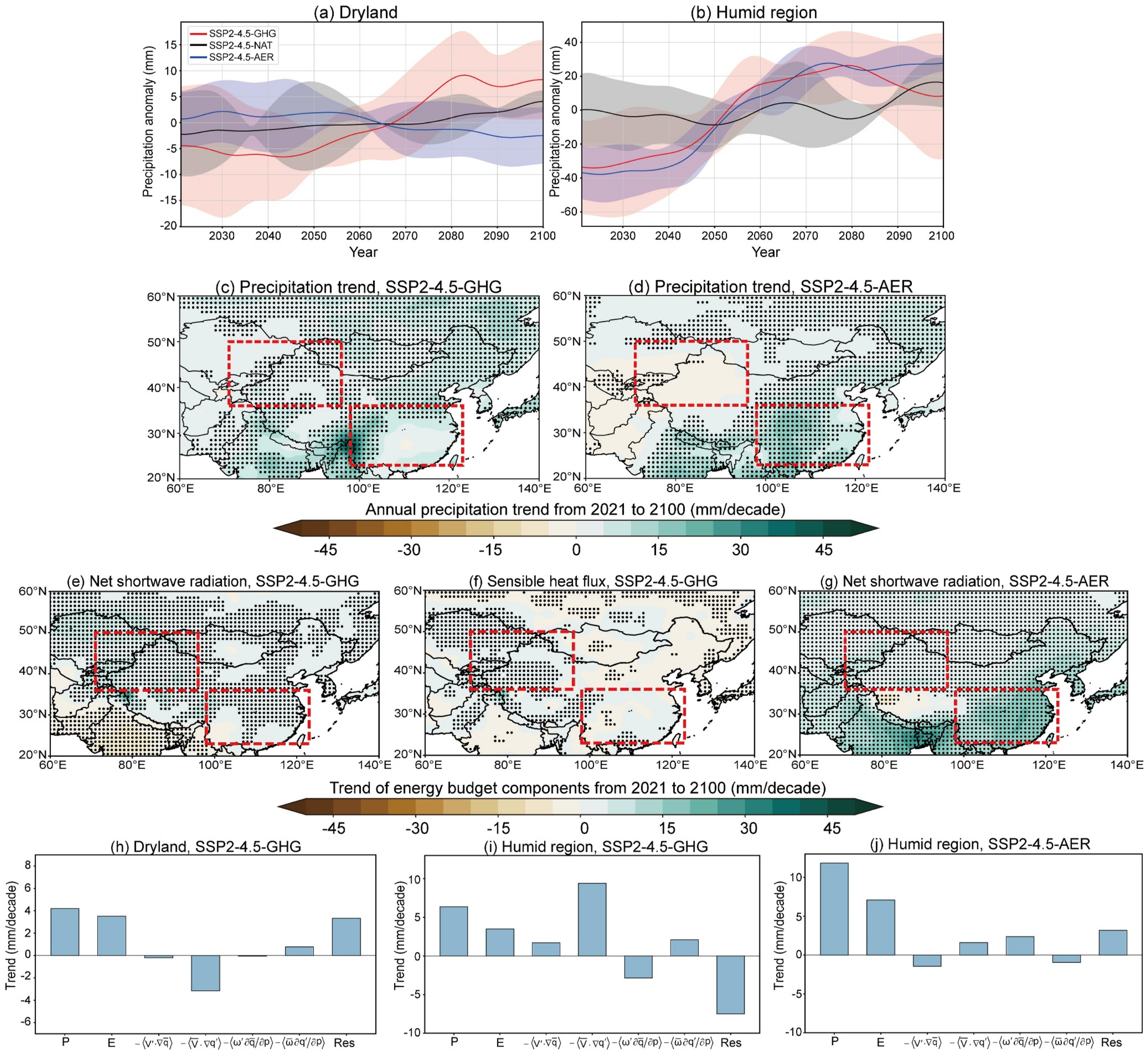

Figure 4. (a), (b) Time series of annual precipitation anomalies in dryland and humid region, respectively, based on the multimodel ensemble mean of different CMIP6 single-forcing experiments under the SSP2-4.5 scenario for the period from 2021 to 2100. Anomalies for each model are calculated with respect to the 2021 to 2100 average. The shaded area is an uncertainty range of one standard deviation across different models. SSP2-4.5-NAT represents natural forcing. (c), (d) Linear trends in annual precipitation under greenhouse gas forcing (SSP2-4.5-GHG) and aerosol forcing (SSP2-4.5-AER) for the period from 2021 to 2100, respectively. The dots indicate significant trends at the 95% confidence level based on the two-tailed Student's *t*-test. (e)–(g) Linear trends of net shortwave radiation and sensible heat flux in the SSP2-4.5-GHG experiment and net shortwave radiation in the SSP2-4.5-AER experiment, respectively. The dots indicate significant trends at the 95% confidence level based on the two-tailed Student's *t*-test. (h)–(j) Terms of the moisture budget equation for dryland in the SSP2-4.5-GHG experiment and for humid region in the SSP2-4.5-GHG and SSP2-4.5-AER experiments, respectively. The CEEMDAN method is used to produce the time series in (a) and (b).

the dryland and humid region in East Asia has weakened. This conclusion is further corroborated by the analysis of precipitation anomaly time series shown in Figure S10 in Supporting Information S1, which demonstrates consistent long-term trends in both observations and simulations.

The external forcings responsible for the dry-wet contrast in East Asian precipitation are further investigated using the single-forcing historical simulations from CMIP6. From 1850 to 2014, greenhouse gas forcing dominates precipitation changes in the dryland, showing about a 3.9% increase in annual precipitation anomalies (Figure 2a). Aerosol forcing leads to a 9.0% reduction in precipitation over humid region, while the greenhouse gas forcing shows only a slight increase (Figure 2b). In contrast, natural forcing exerts minimal influence on both regions, indicating a limited role of natural forcing compared to anthropogenic forcings (Figures 2a and 2b). The spatial distribution of trends further supports these findings. In dryland, greenhouse gas emissions drive significant precipitation increases (Figure 2c), with neither aerosol nor natural forcing producing appreciable changes (Figure 2d and Figure S11 in Supporting Information S1). In humid region, precipitation trends exhibit distinct responses to different forcings. Under greenhouse gas forcing, precipitation increases notably across the region (Figure 2c). When only the aerosol forcing is imposed, a significant reduction in precipitation is observed throughout the humid region (Figure 2d). Analysis of the first EOF modes under the three single-forcing experiments further reveals distinct impacts of external forcings on precipitation patterns. Specifically, greenhouse gas forcing increases precipitation in both dryland and humid region, whereas aerosol forcing decreases precipitation in these areas (Figure S7 in Supporting Information S1).

Moisture and energy budget analyses reveal distinct mechanisms driving precipitation changes across regions. In dryland under greenhouse gas forcing, enhanced net longwave radiation significantly increases evaporation through surface energy balance, consequently leading to increased precipitation (Figures 2e and 2g). The influence of sensible heat flux and net shortwave radiation is minimal (Figures S12a and S12b in Supporting Information S1). Notably, while evaporation in dryland is lower than in humid region, it still shows considerable magnitudes to feed local precipitation except for deserts (Figure S13 in Supporting Information S1). The short-distance moisture transport from the southwest and northwest further sustains the water source for surface evaporation of dryland (Hua et al., 2017). As a result, substantial evaporation can still occur over most areas of dryland. The linear trends of individual components in the moisture budget further validate the identified mechanisms of precipitation change (Figure S14 in Supporting Information S1).

In contrast, in humid region under aerosol forcing, a substantial reduction in net shortwave radiation directly diminishes the energy available at the surface, leading to a marked decrease in evaporation (Figure 2f). While the concurrent reduction in sensible heat flux and increase in net longwave radiation tend to enhance evaporation (Figures S12c and S12d in Supporting Information S1), this effect is overwhelmed by the attenuation of net shortwave radiation. This suppression of evaporation ultimately results in a reduction in precipitation (Figure 2h). Under greenhouse gas forcing, the increase in precipitation in humid region is predominantly driven by the thermodynamic component of horizontal advection, which is closely related to increased humidity (Figures S14e and S15 in Supporting Information S1). The thermodynamic component of vertical convection provides a secondary contribution (Figure S15 in Supporting Information S1). However, in this region, the aerosol-induced reduction in precipitation substantially exceeds the greenhouse gas–induced increase. Therefore, we focus primarily on the physical mechanisms responsible for the precipitation decrease under aerosol forcing. The linear trends of various moisture budget components substantiate the proposed explanations for changes in precipitation (Figure S16 in Supporting Information S1).

Overall, the precipitation distribution in East Asia is characterized by lower precipitation in the northwest dryland and higher precipitation in the southeast humid region. During the historical period, dryland has gradually become wetter, while the humid region has become drier, leading to a weakened precipitation contrast between the two regions. Attribution analysis reveals that greenhouse gas emissions are the main driver of increased precipitation in dryland, while aerosol emissions are the key factor behind decreased precipitation in the humid region. Moisture and energy budget analyses further indicate that evaporation changes, driven by changes in energy balance, regulate local moisture supply and thereby influence precipitation trends in both regions.

3.2. Observational Constraints for Simulated Future Precipitation Change

In dryland, historical simulation indicates an increasing trend in precipitation anomaly, with both SSP2-4.5 and SSP5-8.5 scenarios projecting continued increases from 2015 to 2100 (Figure 3a). In the humid region, despite a decreasing trend during the historical period, future projections under both scenarios suggest a pronounced increase in precipitation, indicating a reversal of the historical decline (Figure 3b). Compared with 2015, precipitation is projected to increase by approximately 98 mm under the SSP2-4.5 scenario and by about 176 mm under the SSP5-8.5 scenario by 2100 (Figure 3b). When normalized by temperature changes (Lee et al., 2021), these correspond to 24.5 mm °C^{-1} under the SSP2-4.5 scenario and 27.08 mm °C^{-1} under the SSP5-8.5 scenario. The linear trends in future scenarios further indicate that precipitation across East Asia will show an increasing trend

(Figure S17 in Supporting Information S1). Notably, the wetting in humid region is more pronounced under the SSP5-8.5 scenario than under the SSP2-4.5 scenario.

Climate model simulations alone are insufficient for precisely predicting future precipitation changes (Chai et al., 2022; Eyring et al., 2019). To overcome this limitation, a principal component-based observational constraint approach has been developed to correct model biases (Huang et al., 2016). This method constrains the leading spatiotemporal modes that dominate precipitation trends over East Asia, which are identified through EOF analysis. Through this approach, observational data can calibrate the sensitivity of dominant spatiotemporal mode changes in East Asian precipitation in simulations, enhancing the accuracy of model projections.

In dryland, observation-constrained historical precipitation changes are relatively minor (Figure 3a). Compared to the HIST period under observational constraints, both SSP2-4.5 and SSP5-8.5 scenarios project decreasing trends, with a more pronounced decline under SSP5-8.5. In contrast, the humid region shows an upward trend in precipitation under both future scenarios, reversing the observation-constrained historical decline (Figure 3b). While the direction of change remains consistent with the raw model projections, the magnitude of the increase is slightly reduced after applying observational constraints.

The original model projection indicates a greater increase in precipitation over the humid region than in the dryland, indicating an enhanced dry-wet contrast across East Asia in the future. Since historical simulations show a higher precipitation increase trend than observations (Figure 3a), the application of observational constraints reduces the simulated trend driven by greenhouse gas forcing (Figure 2c), thus weakening the increasing trend in dryland precipitation in the future. Similarly, in the humid region, observational constraints also weaken the decreasing trend attributed to aerosol forcing during the historical period (Figure 2d). As a result, the future precipitation increase due to aerosols reduction also weakens (Figure 3b), as further supported by Figure 4d in the following section. Consequently, observation-constrained results show a decrease in dryland precipitation and an increase in humid region. Both the original model projections and the projections after observational constraints consistently confirm a strengthened dry-wet contrast in East Asia in the future, representing a reversal of the historical pattern.

3.3. Future Precipitation Changes in Different Forcing Experiments

Given the historical increase and predicted future decrease in aerosol emissions (Figure S18 in Supporting Information S1; Lund et al., 2019), we analyze precipitation changes in both the dryland and humid region under single-forcing experiments within the intermediate-emission scenario, in order to directly assess the effect of reversed aerosol trend. In dryland, greenhouse gas forcing enhances precipitation while aerosol forcing decreases it (Figure 4a). In contrast, in the humid region, precipitation increases under all forcing experiments. Among these, aerosol forcing causes the most substantial increase, rising by 9.9%, while greenhouse gas forcing also leads to a notable increase of about 5.0%, both relative to the climatological precipitation (Figure 4b). The trend further confirms that in dryland, greenhouse gas forcing significantly increases precipitation, while aerosol forcing leads to a reduction in precipitation (Figures 4c and 4d). In the humid region, both greenhouse gas and aerosol forcings lead to a significant increase in precipitation. Although natural forcing has a limited overall impact on precipitation in both dryland and humid region, the weak spatial trends further suggest that natural factors exert only a minor influence on atmospheric moisture modulation compared to anthropogenic forcings (Figure S19 in Supporting Information S1).

In dryland, under greenhouse gas forcing, the increase in net shortwave radiation enhances local evaporation (Figure 4e). Although net longwave radiation and sensible heat flux tend to suppress evaporation, their offsetting effects are weaker than the enhancement induced by net shortwave radiation (Figure 4f and Figure S20a in Supporting Information S1). Consequently, evaporation intensifies in this region, ultimately leading to increased precipitation (Figure 4h). In the humid region, greenhouse gas-induced precipitation increases are primarily driven by the thermodynamic component of horizontal advection, while local evaporation plays a secondary role (Figure 4i). The horizontal advection is primarily driven by moisture transport from the Bay of Bengal and the Arabian Sea, where vertically integrated specific humidity shows a pronounced increasing trend (Figure S21a in Supporting Information S1). To quantify the relative contributions of humidity and circulation changes, we decompose the horizontal moisture flux trend into dynamic (wind-driven) and thermodynamic (humidity-driven) components (Figures S21b and S21c in Supporting Information S1). The results indicate that the thermodynamic component dominates the spatial pattern of the total moisture flux trend. In contrast, the dynamic component is weaker. Changes in the wind field partially inhibit moisture transport, resulting in a direction opposite to that of the total flux trend. Thus, the precipitation increase in humid region is primarily governed by thermodynamic processes, with enhanced horizontal moisture transport resulting mainly from increased atmospheric moisture content rather than significant circulation changes.

Notably, for aerosol-induced precipitation increases in humid region, evaporation is the primary driver, accounting for 60% of the total precipitation change (Figure 4j). The increase in evaporation is primarily attributed to the enhancement of net shortwave radiation, which leads to greater surface solar energy input and thus promotes stronger evaporation (Figure 4g). The impacts of sensible heat flux and net longwave radiation are relatively minor (Figures S20b and S20c in Supporting Information S1). Consequently, future reductions in aerosol emissions are likely to enhance moisture sourced from local evaporation, playing a pivotal role in increasing precipitation in these areas. The spatial patterns of trends in each moisture budget component further validate these findings (Figures S22 and S23 in Supporting Information S1).

Overall, greenhouse gases are the dominant driver of precipitation changes in dryland, mainly by increasing net shortwave radiation, which enhances local evaporation and thereby promotes increased precipitation. In contrast, aerosols are the key determinants of precipitation changes in humid region, mainly by increasing net shortwave radiation, which leads to increased precipitation through enhanced local evaporation. While aerosols are the primary drivers, greenhouse gases still exert a significant influence on humid region, primarily through the thermodynamic component of horizontal advection driven by increased humidity. In future climate scenarios, the impacts of greenhouse gases and aerosols on precipitation in humid region tend to increase, which contrasts with the historical period when aerosols led to decreased precipitation while greenhouse gases contributed to increases (Figures 2b and 4b).

With the anticipated reduction in aerosol emissions, humid region is projected to shift from historical precipitation declines to future increases under aerosol forcing. This transition is primarily attributed to the reversal of net shortwave radiation following aerosol reduction, shifting local evaporation from a historical decline to an increase in the future, thereby enhancing moisture supply and drives precipitation increases. This shift not only signals a reversal in the precipitation trend but also underscores the potential of a reversed dry-wet contrast in East Asia. These results offer crucial insights for predicting future climate patterns and formulating effective adaptation strategies in East Asia.

4. Conclusions

The analysis based on observations and CMIP6 multimodel simulations reveals distinct effects and mechanisms by which greenhouse gases and aerosols modulate precipitation patterns in East Asia, with model-simulated precipitation adjusted using an observation-constrained approach. Historically, greenhouse gases have enhanced evaporation primarily through increased net longwave radiation, thereby contributing to greater precipitation in dryland. In contrast, aerosols have suppressed evaporation by reducing net shortwave radiation, thereby contributing to decreased precipitation in humid region. Under both intermediate- and high-emission scenarios, unconstrained CMIP6 models project increases in precipitation across both regions, with a more pronounced rise in humid region, which suggests a future intensification of the dry-wet contrast. When constrained by observations, the projections indicate a slight decline in dryland precipitation and an increase in humid region. Therefore, both the raw model projections and the observation-constrained projections consistently indicate that the dry-wet contrast in East Asia is likely to strengthen in the future.

Historically, aerosol-increase-induced drying in humid region and greenhouse-gas-increase-driven wetting in dryland weakened the regional precipitation contrast. In contrast, projected future reductions in aerosol emissions are expected to reverse this pattern, as the influence of aerosols shifts from suppressing to enhancing precipitation, particularly in humid region. This reversal is mainly attributed to the shift in net shortwave radiation, which transforms the role of evaporation from a factor that suppressed precipitation in the historical period to one that enhances it under future climate scenarios.

These findings carry significant implications for East Asian ecosystems and human societies, with potential to influence various aspects of regional environmental and socioeconomic dynamics. The historical weakening and projected future strengthening of the dry–wet contrast could reshape regional precipitation patterns, disrupt water resource management, and impact agricultural productivity. Moreover, such shifts may intensify the frequency and intensity of extreme climate events, including droughts and floods (Ranasinghe et al., 2021), thereby challenging the adaptive capacity of local communities and national climate policies.

Our findings are consistent with previous studies showing that reducing aerosol emissions not only improves air quality but also significantly influences regional and global climate patterns. For example, recent studies demonstrate that decreased aerosol emissions in China contribute to an increase in regional precipitation (Jia & Quaas, 2023; Zheng et al., 2020). However, despite CMIP6's extensive exploration of aerosol emission uncertainties through multiple scenarios and models (Scannell et al., 2019), substantial uncertainties persist regarding future aerosol trends. These uncertainties stem from the intricate and unpredictable characteristics of aerosol sources, emission intensities, and atmospheric interactions (Wilcox et al., 2023). Aerosols affect precipitation through microphysical and dynamical processes, with considerable uncertainties compounded from variations in aerosol types and model parameterizations (Rosenfeld et al., 2008). Consequently, current climate models struggle to simulate accurately aerosol impacts on precipitation at both regional and global scales. This limitation stems from insufficient understanding of aerosol effects on small-scale precipitation, biased parameterizations (Malavelle et al., 2017; Stier et al., 2024; Tao et al., 2012), and model biases in representing aerosol-induced changes in large-scale circulations (Bollasina et al., 2011). Although observational constraints can refine model projections, improving the fundamental physical and dynamic representation within climate models remains essential.

Although the observational constraint method employed in this study significantly improves the reliability of model projections, its accuracy is limited by two main factors. First, it depends on the statistical relationship between observational data and model simulations, as noted by Huang et al. (2016). Second, its effectiveness is contingent upon the future persistence of the dominant EOF modes. To further advance understanding and predictive capability, future research should prioritize the following directions: (a) exploring the complex interactions between aerosol types and precipitation processes, (b) quantifying the differential impacts of various aerosol components on the reversal of East Asia's dry-wet contrast, and (c) developing more sophisticated observational constraint techniques that integrate physical processes. Addressing these aspects will enhance the reliability of climate projections and support more informed climate adaptation strategies in East Asia.

Data Availability Statement

CRU monthly precipitation data (Harris et al., 2020) is available at https://crudata.uea.ac.uk/cru/data/hrg/#current. PREC/L monthly precipitation data (Chen et al., 2002) is available at https://psl.noaa.gov/data/gridded/data.precl.html. ERA5 monthly precipitation data (Soci et al., 2024) is available at https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels-monthly-means?tab=overview. CMIP6 monthly precipitation data (Gillett et al., 2016; O'Neill et al., 2017) is available at https://esgf-node.ipsl.upmc.fr/search/cmip6-ipsl/. Code (Zhao, 2025) is available at https://doi.org/10.5281/zenodo.15655695.

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